CACTUS FALL 2026 · WEEK 3 · LEARNING SESSION

Verify and analyze

A number can be computed correctly and still be wrong for the question.
Today you learn to test it.

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THE IDEA

Last week we checked cleaning. This week we check statistics.

Cleaning code

Loses rows or changes values. A row count can reveal it.

Statistics code

Gives one number that looks fine. Nothing on the screen tells you it is wrong.

So we test it

Against an answer we already know.

CONCEPT

What an estimate claims

A difference in means

The average of one group minus the average of the other.

A regression coefficient

With one yes-or-no predictor, the same difference, plus a standard way to say how uncertain it is.

An interval

A range that shows how much the estimate would move if we ran the study again.

Today's example, made up: a platform adds an “AI-generated” label to some posts. Does it change how likely people say they are to share them?

LIVE DEMO 1

One simulated study, with a known answer

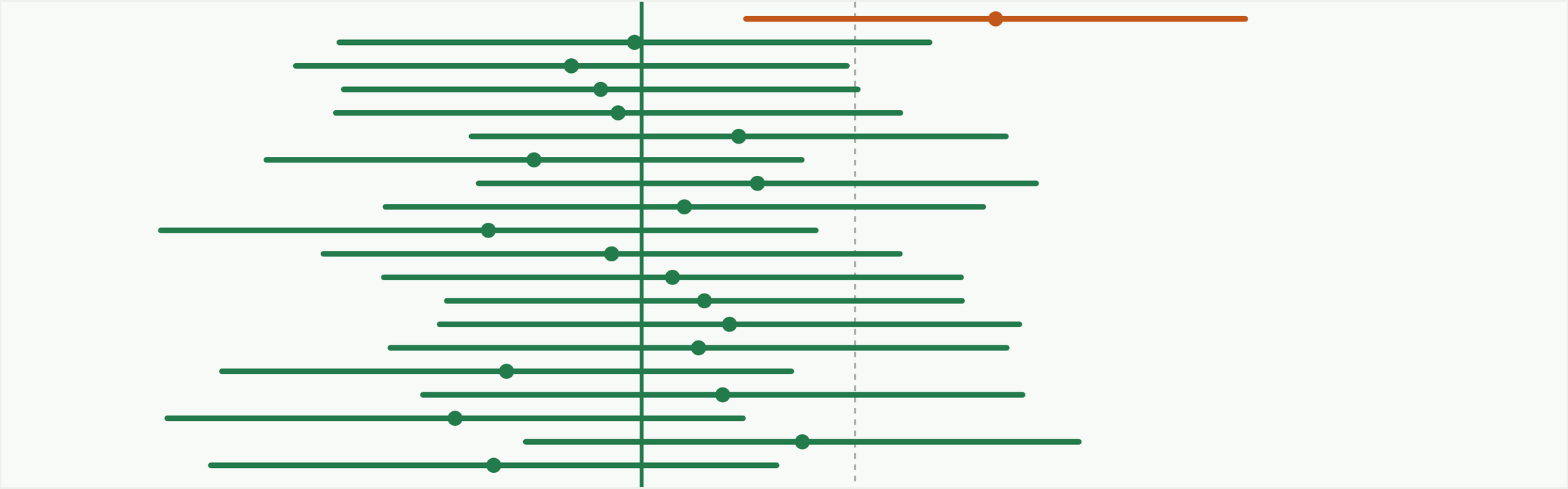
```
d = simulate(100, true_effect=-0.6)
est, lo, hi = estimate(d)
```

True effect	Your estimate	Interval
−0.6	**?**	**? to ?**

Predict first: will the estimate be close to −0.6? Will it be exactly −0.6?

NOISE AND SAMPLE SIZE

Run it again, and the estimate moves



Twenty studies of 100 people each, with the same true effect of −0.6 (solid green line; dashed line is zero). Each line is one study's interval. **Orange is an interval that misses the truth.**

CONCEPT

What an interval says, and what it does not

It says

If we ran the study many times, about 95 of 100 intervals built this way would contain the true effect.

It does not say

That the true effect has a 95% chance of being inside this one interval. Or that a result below .05 is a verdict.

A wider interval means more uncertainty. A bigger sample makes it narrower.

CONCEPT

A confounder can flip the answer

Example

Students who attend tutoring score **lower** than students who do not.

The confounder

Students who were already struggling are more likely to attend tutoring.

What it does

The comparison mixes the effect of tutoring with who chooses tutoring.

An experiment avoids this by deciding who gets the treatment by chance. **Survey or platform data** usually cannot. You will see this in the coding session.

LIVE DEMO 2

Check an AI’s answer against the truth

```
Using this table, estimate the effect of the label on sharing intent and tell me what
it shows. (paste columns and sample rows)
```

1. Compare its number with the known effect, −0.6

2. Read what it says the result proves

3. Ask: what could I not tell from this answer?

WRITING IT UP

Over-claiming

A claim you will see	What it needs
“The label causes lower sharing.”	A design that supports a cause, such as random assignment
“It is significant, so it is large.”	Look at the size of the estimate, not only at the p-value
“No effect was found, so there is none.”	A small sample may be too noisy to tell
“The AI said so.”	A check you ran yourself

A FIRST LOOK

A notebook belongs in a repository

History

Every saved version is kept. You can see what changed.

Sharing

A link lets a supervisor or a co-author open exactly what you ran.

Never include

Keys, passwords, or private data. Anything you commit may be visible.

You will do this in the coding session. Create a free GitHub account before then.

WHAT TO DO NEXT

Before the coding session

Practice page

Planted-effect lab, about 20 minutes. It ends with a small Colab task.

Required reading

Textbook Chapter 8, sections 8.1 and 8.2, and Chapter 4, section 4.3.

Set up

A free GitHub account, and a working key or a clear plan for Week 4 data access.

In the coding session: simulate, recover the effect, find three bugs in AI code, and see a confounder flip an answer.